Part 1 – Build Flask Docker Image
1. Clone the repository

Clone the Flask app repository to your local machine

```
mohamed@DESKTOP-N92AEE4:~/containers$ git clone https://github.com/meldafrawi/basic-flask-app.git
Cloning into 'basic-flask-app'...
remote: Enumerating objects: 32, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 32 (delta 8), reused 8 (delta 8), pack-reused 11 (from 1)
Receiving objects: 100% (32/32), 274.91 KiB | 1.14 MiB/s, done.
Resolving deltas: 100% (8/8), done.
mohamed@DESKTOP-N92AEE4:~/containers$ cd basic-flask-app/
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ touch Dockerfile
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ vim Dockerfile
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$
```

2. Create a Dockerfile

Create a Dockerfile in the project folder

```
FROM python:3.6-alpine

WORKDIR /app

COPY . .

RUN pip3 install -r requirements.txt

EXPOSE 5000

CMD ["python", "routes.py"]
```

Build the Docker image

an Error of requirements compatability

```
=> ERROR [4/4] RUN pip install --no-cache-dir -r requirements.txt
------
 > [4/4] RUN pip install --no-cache-dir -r requirements.txt:
23.11 Collecting appdirs==1.4.3 (from -r requirements.txt (line 1))
23.48   Downloading appdirs-1.4.3-py2.py3-none-any.whl.metadata (8.8 kB)
23.57 Collecting click==6.7 (from -r requirements.txt (line 2))
23.63   Downloading click-6.7-py2.py3-none-any.whl.metadata (424 bytes)
23.72 Collecting Flask==0.12 (from -r requirements.txt (line 3))
23.79   Downloading Flask-0.12-py2.py3-none-any.whl.metadata (1.9 kB)
23.86 Collecting itsdangerous==0.24 (from -r requirements.txt (line 4))
23.92   Downloading itsdangerous-0.24.tar.gz (46 kB)
24.01     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.5/46.5 kB 466.5 kB/s eta 0:00:00
24.06   Preparing metadata (setup.py): started
23.34   Preparing metadata (setup.py): finished with status 'done'
23.42 Collecting Jinja2==2.9.5 (from -r requirements.txt (line 5))
23.49   Downloading Jinja2-2.9.5-py2.py3-none-any.whl.metadata (2.2 kB)
23.69 Collecting MarkupSafe==1.0 (from -r requirements.txt (line 6))
23.76   Downloading MarkupSafe-1.0.tar.gz (14 kB)
23.77   Preparing metadata (setup.py): started
23.99   Preparing metadata (setup.py): finished with status 'error'
24.00   error: subprocess-exited-with-error
24.00
24.00   × python setup.py egg_info did not run successfully.
24.00   │ exit code: 1
24.00   ╰─> [6 lines of output]
24.00       Traceback (most recent call last):
24.00         File "<string>", line 2, in <module>
24.00         File "<pip-setuptools-caller>", line 34, in <module>
24.00         File "/tmp/pip-install-w18ijlph/markupsafe_107e246acab245699d59cf5c9c9b3fff/setup.py", line 6, in <module>
24.00           from setuptools import setup, Extension, Feature
24.00       ImportError: cannot import name 'Feature' from 'setuptools' (/usr/local/lib/python3.11/site-packages/setuptools/__init__.py)
24.00       [end of output]
24.00
24.00   note: This error originates from a subprocess, and is likely not a problem with pip.
24.00 error: metadata-generation-failed
24.00
24.00 × Encountered error while generating package metadata.
24.00 ╰─> See above for output.
24.00
```

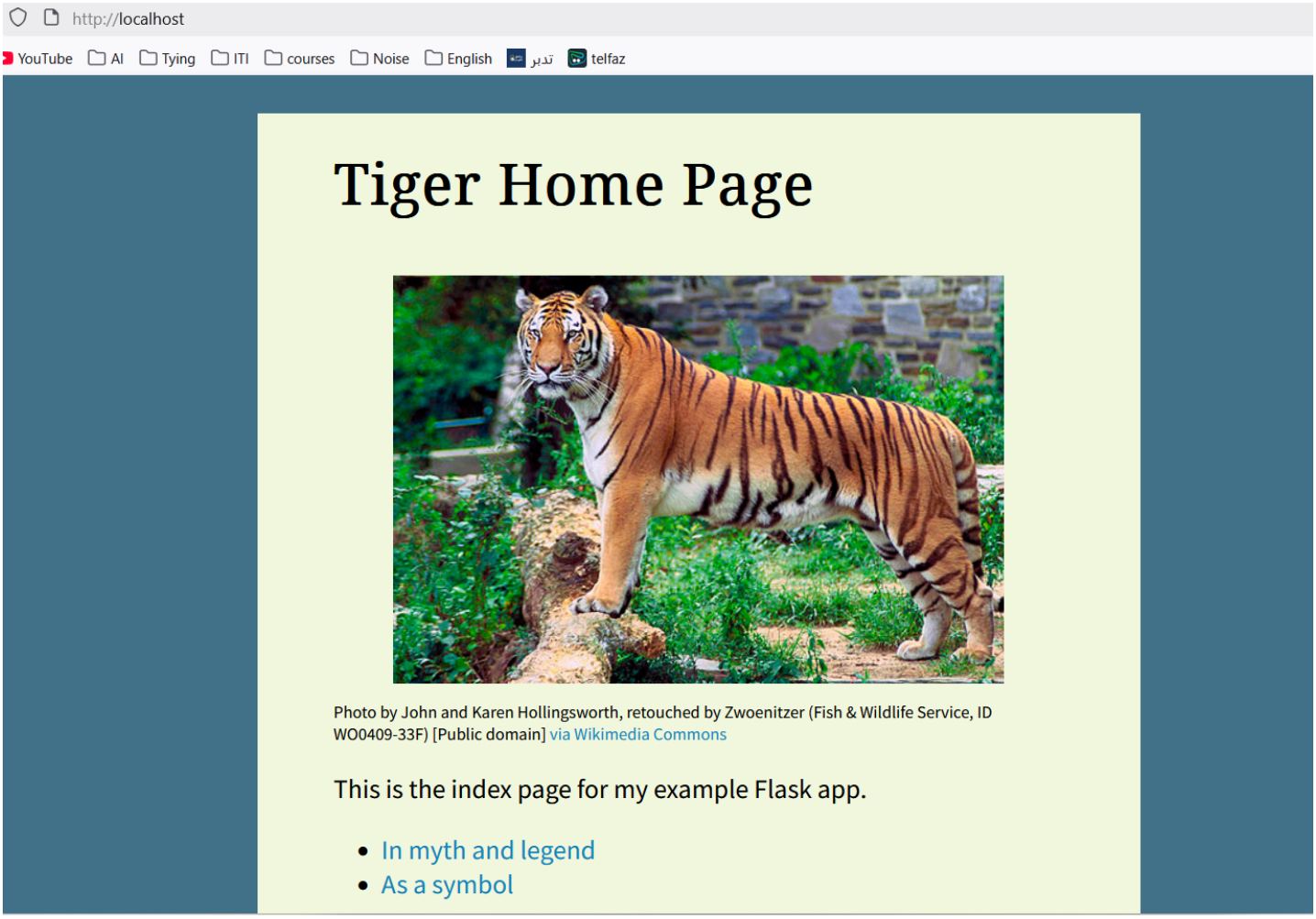
I installed flask instead of requirements.text

```
FROM python:3.13-alpine

WORKDIR /app

COPY . .

RUN pip install flask

EXPOSE 5000

CMD ["python", "routes.py"]
```

successfully built

```
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ docker build -t iti-flask-lab2 .
 => => unpacking to docker.io/library/iti-flask-lab2:latest
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ docker images

IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
iti-flask-lab2:latest    19e9c353e5ac       93.1MB         23.1MB
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$
```

run



login to docker hub
tag the image
push image to docker hub

```
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ docker login
Authenticating with existing credentials... [Username: mohammedmostafanada]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ docker tag iti-flask-lab2 mohammedmostafanada/iti-flask-lab2
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$ docker push mohammedmostafanada/iti-flask-lab2
Using default tag: latest
The push refers to repository [docker.io/mohammedmostafanada/iti-flask-lab2]
98914020a3be: Pushed
a769d56b7aba: Pushed
79a000d511eb: Pushed
e965247e4f6f: Pushed
02b83fe93be9: Pushed
3ce7782530be: Pushed
be2fee0bf4ec: Pushed
589002ba0eae: Pushed
latest: digest: sha256:19e9c353e5aca8e0329fae5f6b5af25f632fd8b3c3367fb2e99b30ed09cc1f38 size: 856
mohamed@DESKTOP-N92AEE4:~/containers/basic-flask-app$
```

image link:

https://hub.docker.com/r/mohammedmostafanada/iti-flask-lab2

Part-2

create iti-network: 10.0.0.0/8 (bridge)

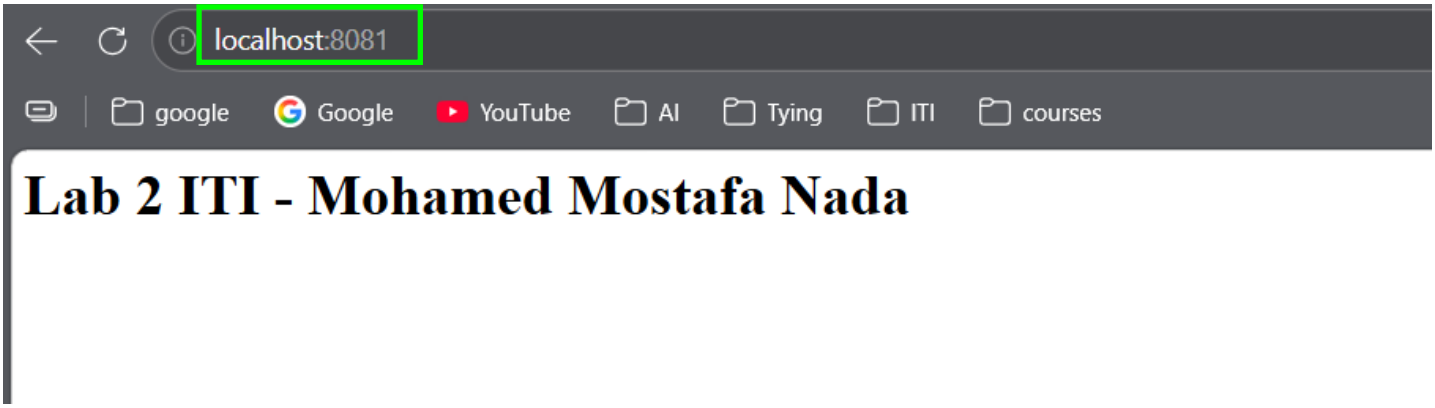
```
mohamed@DESKTOP-N92AEE4:~/containers$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
49721ab4a381   bridge    bridge    local
c9df3589c275   host      host      local
f503cb024254   none      null      local
mohamed@DESKTOP-N92AEE4:~/containers$ docker network create --driver bridge --subnet 10.0.0.0/8 iti-network
6f307ec602a95d51f348cb4c7257ef8a3dd3ab190d5b0a82ead07b09ef402b52
mohamed@DESKTOP-N92AEE4:~/containers$ docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
49721ab4a381   bridge        bridge    local
c9df3589c275   host          host      local
6f307ec602a9   iti-network   bridge    local
f503cb024254   none          null      local
mohamed@DESKTOP-N92AEE4:~/containers$
```

run webserver-iti container

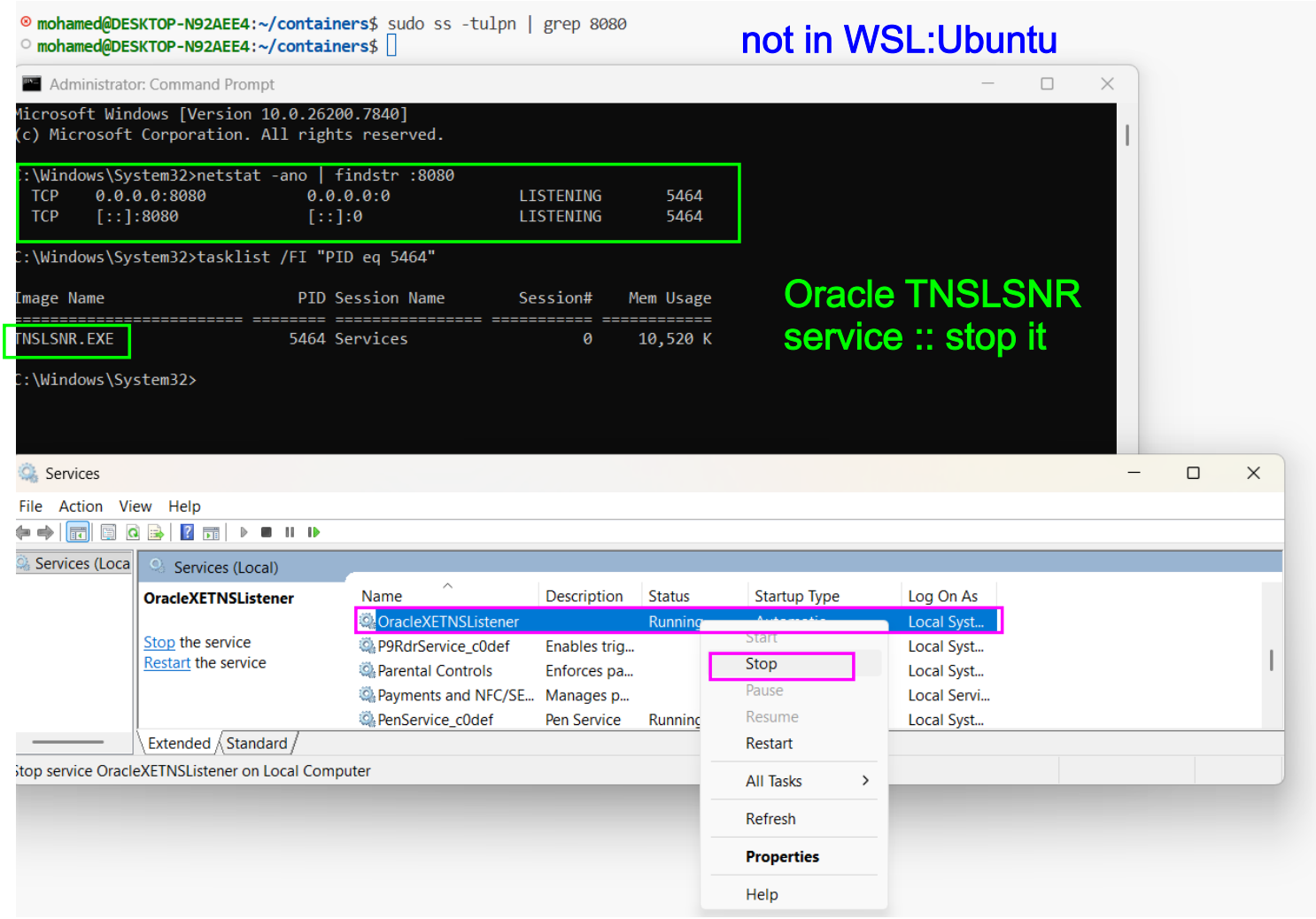
error: port 8080 is in use

changed the port and run again it worked successfully

```
mohamed@DESKTOP-N92AEE4:~/containers$ docker run -d --name webserver-iti --network iti-network -p 8080:8080 -v /home/mohamed/containers/html:/usr/share/nginx/html nginx:alpine
295c2a58f17a2f8b613a2aad2c1848af36dc3ddb4730c30255aa17e3d45dc7a7
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 -> 127.0.0.1:0: /forwards/expose returned unexpected status: 500

Run 'docker run --help' for more information
mohamed@DESKTOP-N92AEE4:~/containers$ docker rm -f webserver-iti
webserver-iti
mohamed@DESKTOP-N92AEE4:~/containers$ docker run -d --name webserver-iti --network iti-network -p 8081:80 -v /home/mohamed/containers/html:/usr/share/nginx/html nginx:alpine
c1120bf76065937a2b3eba72d68317f28c55ad7ba69e6b4e326019eb6a0f319e
mohamed@DESKTOP-N92AEE4:~/containers$
```



tried to know which service is using port 8080

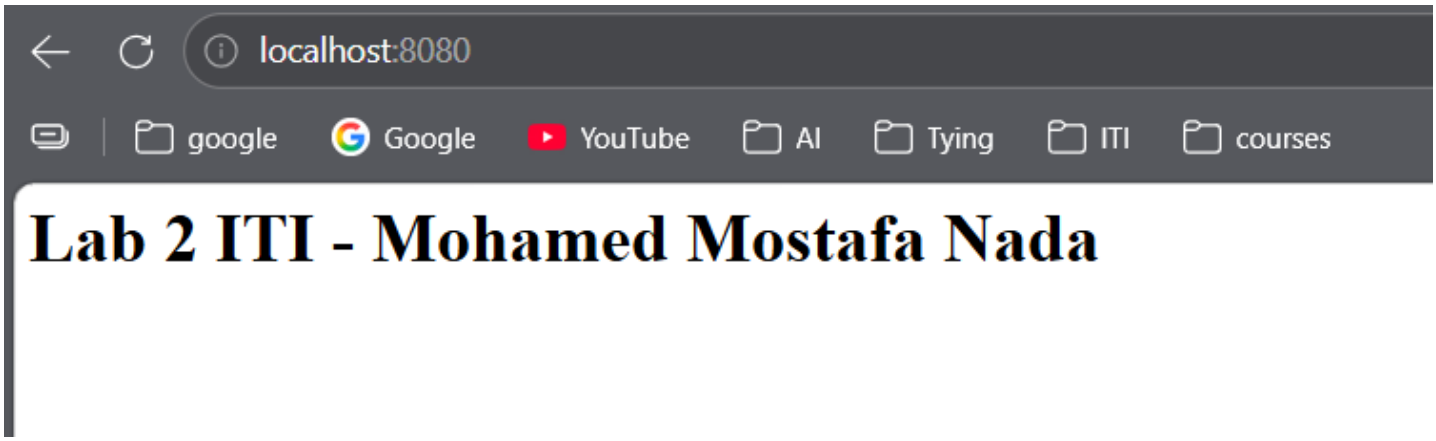


rerun the container on port 8080:8080

```
mohamed@DESKTOP-N92AEE4:~/containers$ docker rm -f webserver-iti
webserver-iti
mohamed@DESKTOP-N92AEE4:~/containers$ docker run -d --name webserver-iti --network iti-network -p 8080:8080 -v /home/mohamed/containers/html:/usr/share/nginx/ht
ml nginx:alpine
877eae24bb576b7219ccb44101c7e574b8838b45ca357a5f593e7e9de3e4fad3
mohamed@DESKTOP-N92AEE4:~/containers$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
877eae24bb57   nginx:alpine   "/docker-entrypoint.…"   20 seconds ago   Up 19 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   webserver-iti
mohamed@DESKTOP-N92AEE4:~/containers$
```

nginx listen by default on port 80 >> changing it form default.conf file

```
mohamed@DESKTOP-N92AEE4:~/containers$ docker exec -it webserver-iti sh
/ #  vi /etc/nginx/conf.d/default.conf
/ # nginx -s reload
2026/03/03 23:14:11 [notice] 49#49: signal process started
/ #
```

```
server {
    listen       8080;
    listen  [::]:8080;
    server_name  localhost;
```

part2 completed successfully :)